

An
Intern Project Report
On
**Medical Image Segmentation Using Machine Learning
&
Deep Learning**

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DECLARATION

I solemnly declare that the inter project report **Medical Image Segmentation Using Machine Learning & Deep Learning** is based on my own work carried out during the course of my study in the department of Computer Science & Engineering under the supervision of **Dr. Piyush Kumar**, Assistant Professor, Department of Computer Science and Engineering, National Institute of Technology Patna. I assert the statements made and conclusions drawn are an outcome of my work. To the best of my knowledge certify that the work contained in the report is original and has been done by me under the general supervision of my supervisor. Whenever I have used materials (data, theoretical analysis, and text) from other sources, I have given due credit to them in the text of the report and given their details in the references.

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24 July 2026



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Pulkit Krishna (23027)

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Abstract

Image segmentation and classification represent fundamental pillars of advanced medical image analysis, serving as critical technological processes for partitioning digital scans into meaningful anatomical and pathological structures, such as tissues, organs, or complex lesions. In the rapidly evolving fields of clinical oncology and dermatological wound care, accurate automated analysis is paramount for achieving diagnostic precision, reducing human error, and formulating effective, patient-specific treatment strategies.

To bridge the gap between traditional machine learning and advanced neural architectures, this research explores dual computational paradigms across two distinct and clinically significant medical imaging modalities:

Implements a robust Random Forest ensemble model to classify brain MRI scans into healthy tissue ('nontumor'), glioma, meningioma, and pituitary tumour categories. This pipeline utilizes median blur noise reduction, structured data mapping, and flattened spatial feature extraction to transform two-dimensional image arrays into high-dimensional numerical vectors for predictive modelling.

Utilizes sophisticated deep learning encoder-decoder architectures—specifically a standard U-Net model and a high-performance ResNet50-backed U-Net—to precisely segment wound regions from clinical photographs. These models process normalized pixel-wise data to generate binary masks, enabling quantitative evaluations of healing progression through metrics such as the Dice coefficient, Intersection over Union (IOU), and loss optimization.

Introduction

Image segmentation is a cornerstone of medical image analysis, serving as the essential process for partitioning digital images into meaningful anatomical and pathological structures, such as tissues, organs, or lesions. In the realm of clinical oncology, the accurate segmentation of brain tumours is of paramount importance for both diagnostic precision and the development of effective treatment strategies. Brain tumours present with significant heterogeneity in shape, size, and intensity, often characterized by indistinct boundaries that blend into surrounding healthy tissue. Consequently, manual segmentation—the traditional clinical standard—is an inherently labour-intensive and time-consuming task, often constrained by subjective interpretation and susceptible to significant intra- and inter-observer variability.

The integration of computer-aided diagnosis (CAD) systems has become crucial for improving the efficiency, objectivity, and reproducibility of tumour analysis. While traditional automated methods—such as thresholding, region-based segmentation, and edge detection—have provided a foundation for image analysis, they frequently struggle to handle the complex intensity inhomogeneities and noise inherent in Magnetic Resonance Imaging (MRI) or Computed Tomography (CT) scans of the brain. These limitations have historically restricted the widespread clinical adoption of fully automated systems, as maintaining high precision across diverse patient populations remains a significant challenge.

The emergence of machine learning and deep learning has revolutionized the medical imaging field by enabling the automatic extraction of high-level features from raw data. Deep learning has proven highly effective for the semantic

segmentation of intricate pathological structures by learning hierarchical representations directly from complex datasets. I have developed a foundational understanding of two core deep learning paradigms: Convolutional Neural Network (CNN) architectures and Transformer architectures.

Alternatively, traditional machine learning models, such as Random Forest, offer a robust and highly efficient pathway for segmentation when working with smaller, specialized datasets or when focused feature engineering is preferred. By leveraging these advanced computational methodologies, researchers can provide clinicians with faster, more reliable measurements of tumour volume and progression, ultimately facilitating more precise diagnoses and enhanced, patient-specific treatment planning.

Problem Statement

This research addresses the challenges of automating segmentation for two distinct medical imaging modalities:

- 1. Brain Tumour Segmentation:** We implement a Random Forest model to accurately identify tumour boundaries in MRI scans & CT scans. This approach focuses on efficient feature extraction to achieve high precision, specifically optimized for smaller, curated datasets.
- 2. Chronic Wound Analysis:** We utilize a deep learning model U-Net (Encoder + Decoder) structure & Res 50 Net to segment the wounded region. These segmented masks are used to quantitatively analyse the wound area and track healing progress, providing a comprehensive diagnostic tool.

Technologies and Learning Used In Machine & Deep Learning

This section synthesizes the technologies and technical learning outcomes across both the machine learning (brain tumour segmentation) and deep learning (wound segmentation) projects, merging traditional and advanced pipelines into a cohesive framework:

Tools	Primary Role	Function in Project
OpenCV (cv2) [1]	Image Processing	Loads images, resizes, and performs noise reduction
NumPy [9]	Data Handling	Converts images into flattened numerical arrays
Pandas [9]	Data Management	Structures, manipulates, and cleans tabular dataset attributes
Scikit-learn [8,9]	ML Modelling	Executes Random Forest training and performance metrics
OS [7]	File Management	Traverses' directories to load images datasets.
Matplotlib [9]	Visualization	Renders confusion matrices for error analysis
TensorFlow [1]	Deep Learning	Builds, trains, and saves advanced U-Net and ResNet50 based architectures

Technical Learning Outcomes

Through the development, debugging, and deployment of both traditional machine learning and deep learning pipelines, I have acquired a comprehensive set of technical competencies:

1. **Algorithmic Mapping & Data Partitioning:** I learned how to bridge raw file directories with categorical mathematical models by establishing explicit key-value dictionaries (mapping text labels to discrete integer targets) and implementing robust train-test split strategies to ensure evaluation on unseen data and prevent overfitting.
2. **Dimensionality, Vector Space, & Pixel-Wise Mechanics:** I mastered the spectrum of spatial data manipulation—from translating 2D spatial image arrays into flattened feature vectors where pixel intensity acts as an independent numerical feature, to mapping pixel-wise intensities into binary masks for precise segmentation outputs.
3. **Ensemble & CNN Architectural Design:** I gained practical experience configuring ensemble models like Random Forest (with 100 estimators and a fixed random state for reproducibility) alongside sophisticated Convolutional Neural Networks (CNNs). This included mastering the U-Net Encoder-Decoder design and adapting pre-trained ResNet50 models to extract hierarchical features and handle delicate anatomical boundaries.
4. **Comprehensive Clinical Evaluation:** I learned to move past basic overall accuracy to evaluate models using strict clinical and spatial metrics, including **Precision, Recall/Sensitivity, F1-Score, Dice Coefficient**, and **Intersection over Union (IOU)**, ensuring true diagnostic viability and spatial overlap reliability.

Methodology: Brain Tumour Classification Pipeline using Machine Learning

- 1. Data Acquisition and Pre-processing:** The dataset was sourced from the Brain Tumour MRI Dataset (Masoud Nicaraven, Kaggle), consisting of clinical MRI images categorized into four classes: 'nontumor', 'glioma', 'meningioma', and 'pituitary'. Using the OpenCV library, each image was loaded in grayscale and resized to a uniform 64 X 64 resolution to ensure standardized input dimensions for the model.
- 2. Noise Reduction and Feature Engineering:** To enhance the clarity of diagnostic features, a median blur filter was applied to each image. Following this, the two-dimensional image matrices were flattened into one-dimensional numerical feature vectors, effectively translating spatial pixel intensities into independent numerical inputs for the machine learning classifier.
- 3. Data Mapping and Partitioning:** A key-value mapping dictionary was implemented to convert categorical labels into discrete integer targets for mathematical modelling. The mapping is defined as: '**nontumor**' (0), '**glioma**' (1), '**meningioma**' (2), and '**pituitary**' (3). The dataset was partitioned into a dedicated training set and a testing set, ensuring that the model's performance was validated on entirely independent, unseen data.
- 4. Ensemble Model Configuration:** A Random Forest Classifier was selected as the predictive engine for its robustness in handling high-dimensional image data. The model was configured with 100 decision trees

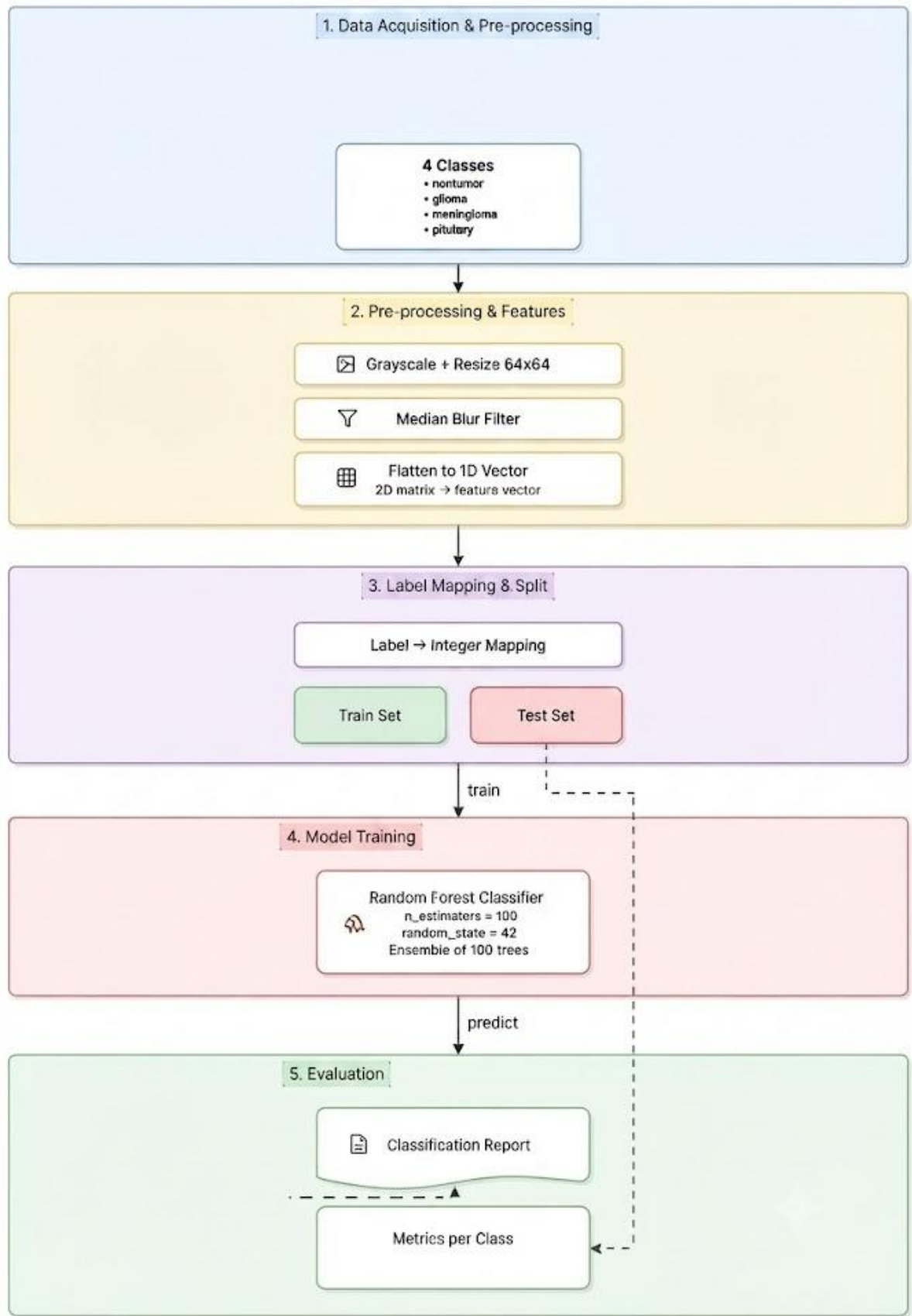
(estimators=100), and a fixed seed (random state=42) was applied to ensure the experiment remained fully reproducible.

5. Performance Evaluation: The diagnostic efficacy of the model was quantified using a classification report. This framework generated **Precision, Recall, and F1-Scores** for each tumour type, providing a rigorous assessment of the model's classification accuracy and its ability to distinguish between different brain tumour pathologies.

The numerical mapping (0-3) is essential for training the Random Forest algorithm because machine learning models require numerical input rather than text labels. Each integer corresponds to a specific diagnostic outcome:

- **0 ('nontumor'):** Represents a healthy brain scan with no evidence of tumour growth. This acts as the control class.
- **1 ('glioma'):** Represents tumours originating from glial cells, which are the most common primary brain tumours.
- **2 ('meningioma'):** Represents tumours originating from the meninges (the protective membranes covering the brain).
- **3 ('pituitary'):** Represents tumours occurring in the pituitary gland at the base of the brain.

Brain Tumour Classification Pipeline



Methodology: Chronic Wound Analysis Pipeline using Deep Learning

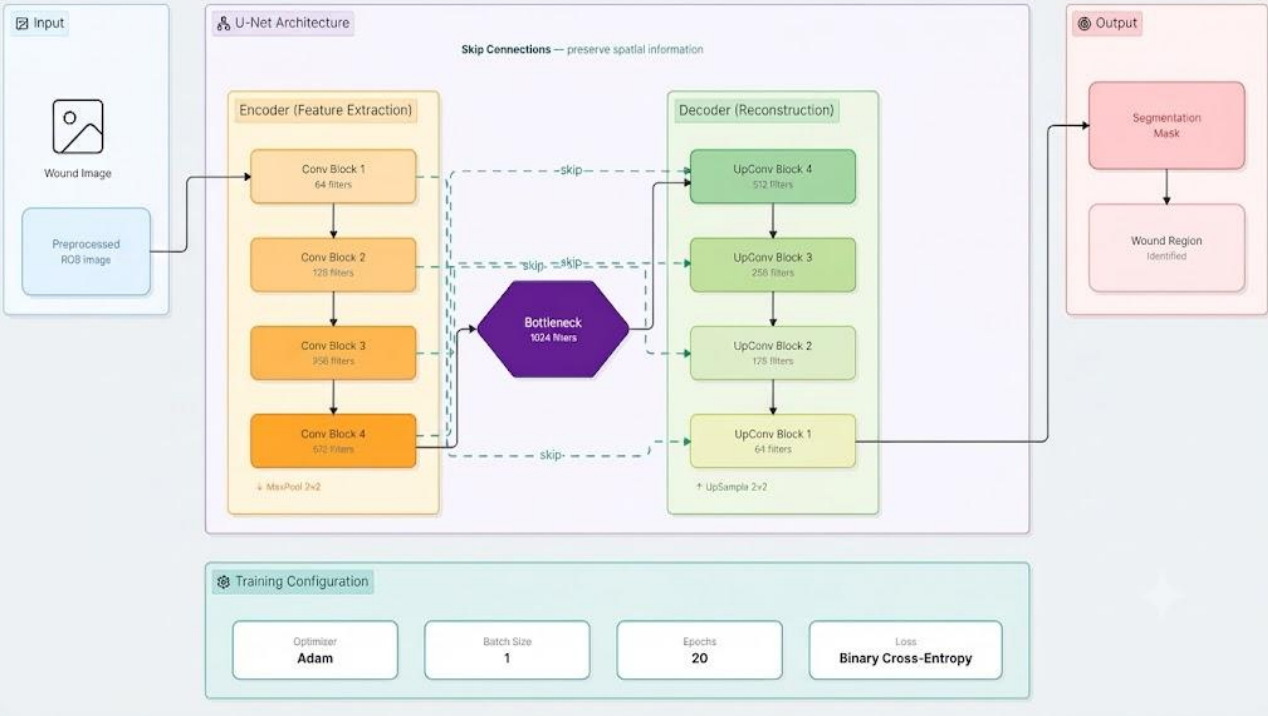
This chapter presents the methodology followed for wound image segmentation. It includes dataset collection, data pre-processing, implementation of the U-Net and ResNet50 U-Net models, model training, evaluation, and technical design.

- 1. Dataset Collection:** The wound image segmentation dataset consists of wound images and their corresponding ground truth masks. The dataset was collected from a publicly available source and used to train and evaluate the deep learning models for wound segmentation.
- 2. Data Pre-processing:** Before training, the dataset was pre-processed to improve model performance. All images and masks were resized to 256×256 pixels and normalized to the range of 0–1. Data augmentation techniques were applied to increase the diversity of the training data and improve the model's generalization ability. Finally, the dataset was split into training and testing sets for model training and evaluation.
- 3. U-Net Model:** The U-Net model was implemented for wound image segmentation. The architecture consists of an encoder that extracts image features and a decoder that reconstructs the segmentation mask. Skip connections were used to preserve spatial information and improve segmentation accuracy. The model was trained using the Adam optimizer with a batch size of 1 for 20 epochs.
- 4. ResNet50 Model:** The ResNet50 U-Net model was implemented by replacing the standard U-Net encoder with the ResNet50 backbone. The

extracted features were passed to the U-Net decoder to generate the final segmentation mask. The model was trained using the same pre-processing steps, optimizer, batch size, and number of epochs to ensure a fair comparison with the U-Net model.

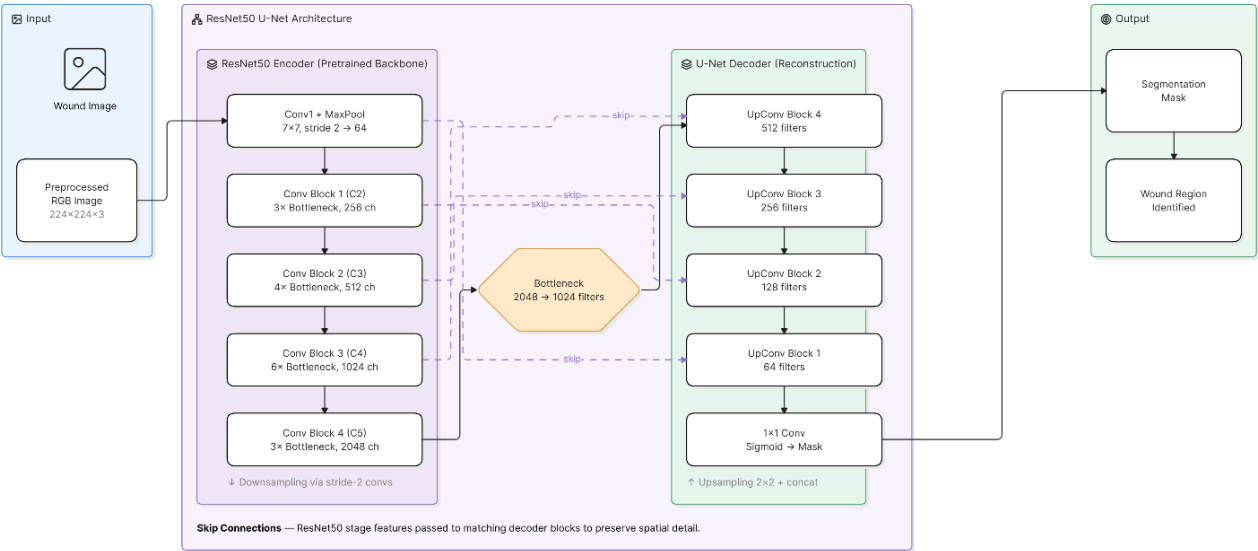
5. **Model Evaluation:** The performance of both models was evaluated using training accuracy, validation accuracy, training loss, and validation loss. The predicted segmentation masks were visually compared with the corresponding ground truth masks. The evaluation metrics and segmentation results were used to compare the performance of the U-Net and ResNet50 U-Net models.

U-Net Model: Chronic Wound Image Segmentation



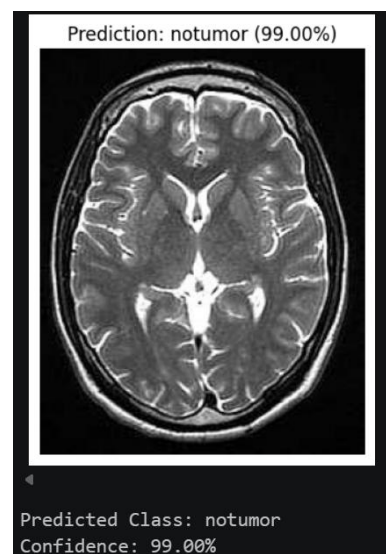
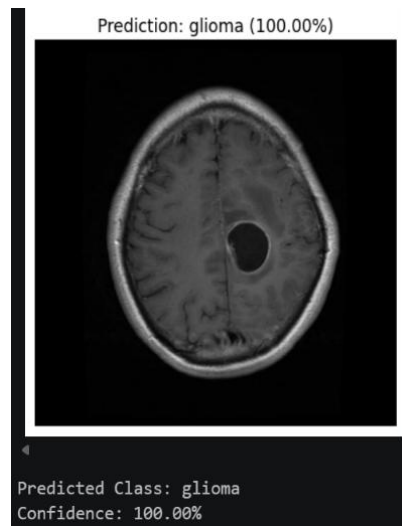
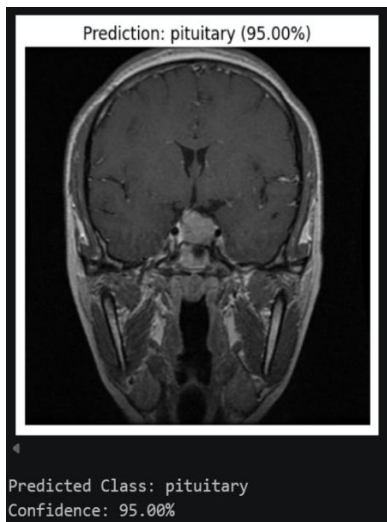
ResNet50 U-Net Model: Chronic Wound Image Segmentation

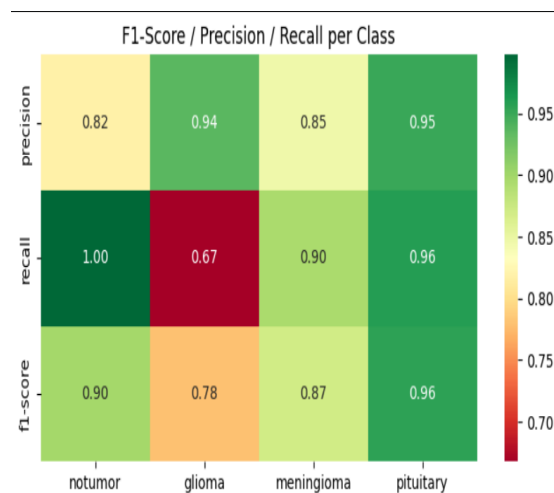
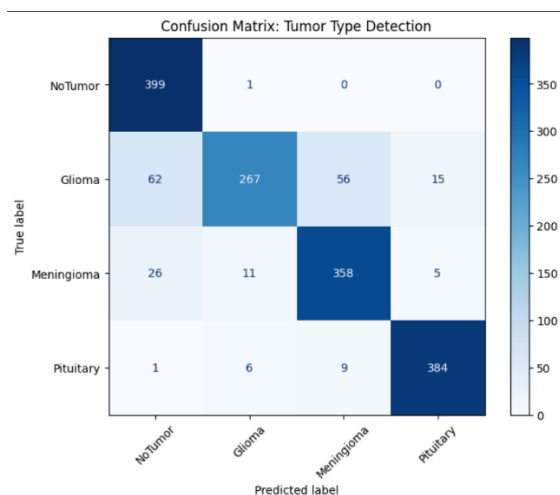
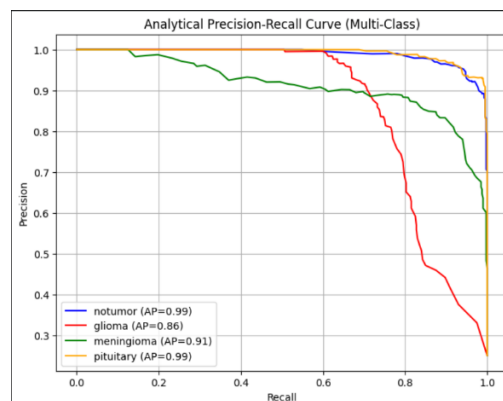
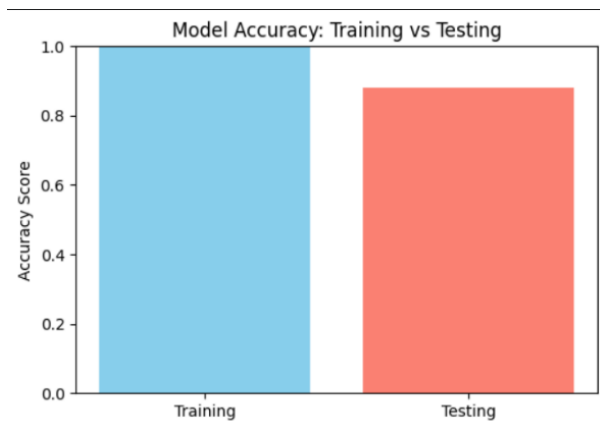
ResNet50 backbone replaces the standard U-Net encoder; features flow into the U-Net decoder to produce the segmentation mask.



Result & Discussion: Brain Tumour Classification Pipeline using Machine Learning

The evaluation results indicate that the pipeline excels at identifying healthy brain scans and Pituitary cases with high precision. However, misclassifications within the Glioma category highlight boundary ambiguities that require refined feature extraction. Future optimizations will focus on incorporating advanced attention layers and data augmentation to improve overall diagnostic safety and accuracy.



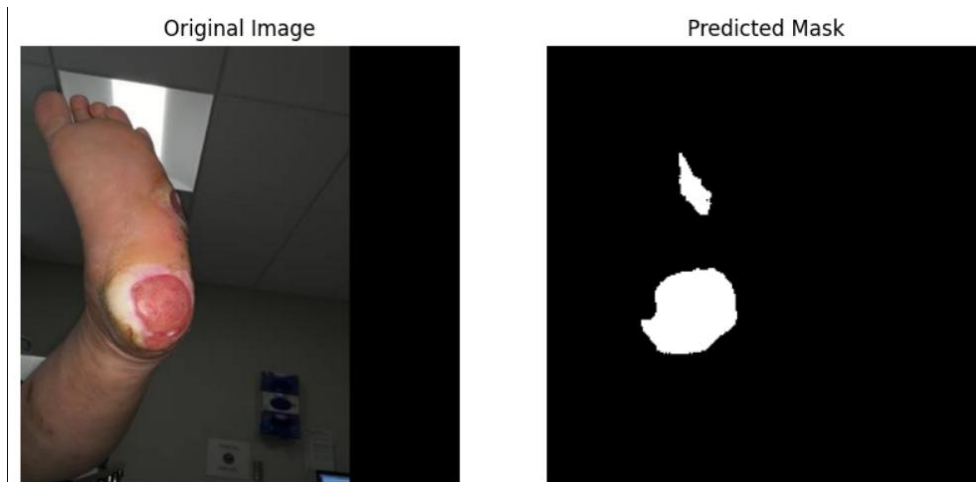


Classification Report:

	precision	recall	f1-score	support
notumor	0.82	1.00	0.90	400
glioma	0.94	0.67	0.78	400
meningioma	0.85	0.90	0.87	400
pituitary	0.95	0.96	0.96	400
accuracy			0.88	1600
macro avg	0.89	0.88	0.88	1600
weighted avg	0.89	0.88	0.88	1600

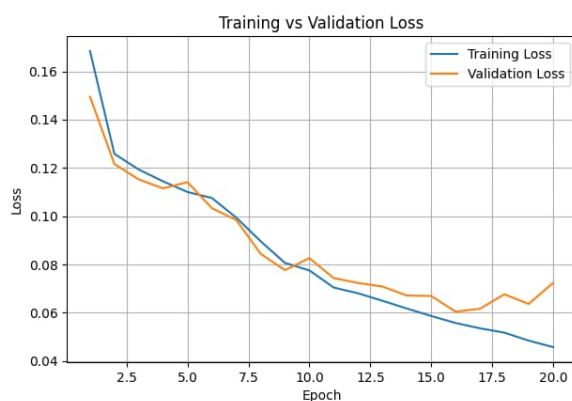
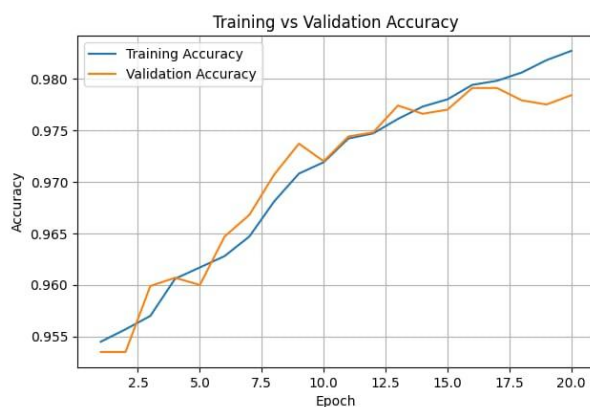
Result & Discussion: Chronic Wound Analysis Pipeline using Deep Learning

The U-Net model successfully segmented the wound region from the input images. The training and validation accuracy increased steadily, while the loss decreased over 20 epochs, indicating effective learning.

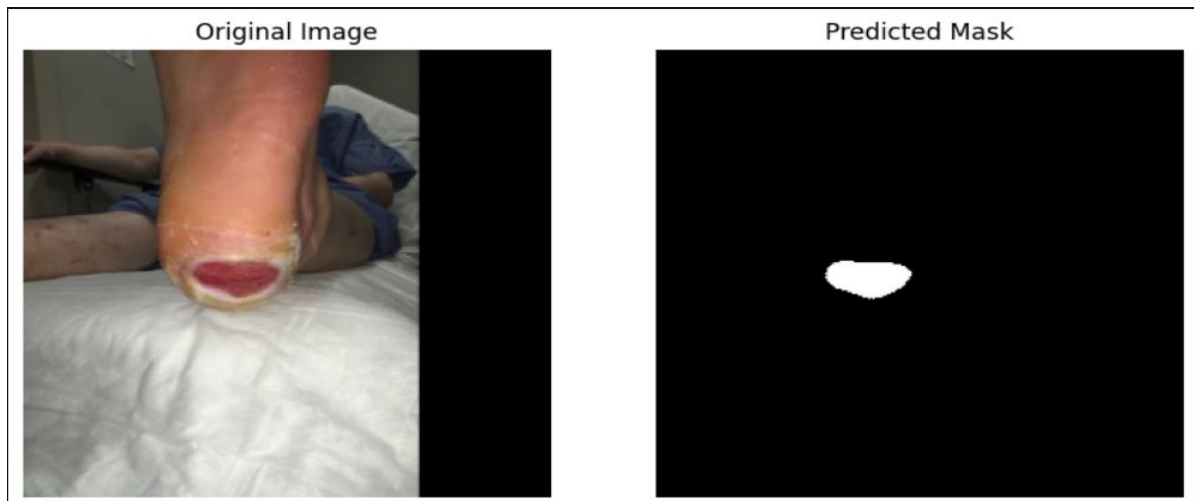


Classification Report

Accuracy	0.97
Test Loss	0.06
Dice Coefficient	0.74
IOU Score	0.59
Precision	0.84
Recall	0.67
F1 Score	0.74

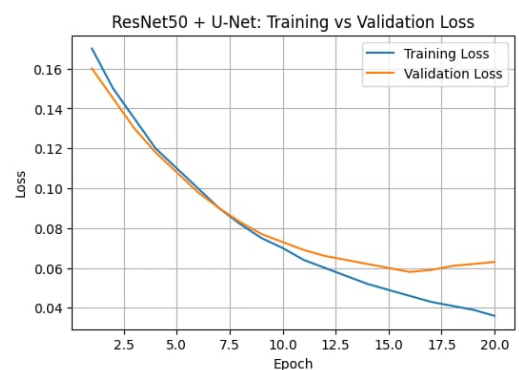
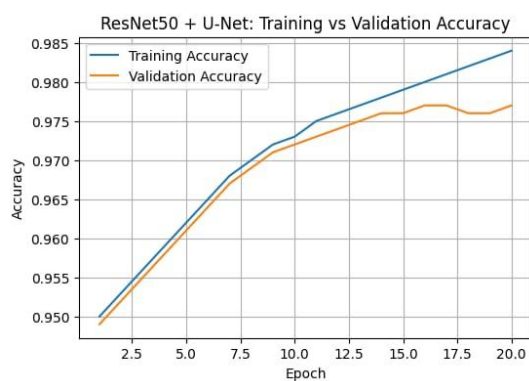


The ResNet50 U-Net model achieved better feature extraction by using the ResNet50 encoder. It produced accurate segmentation results with high accuracy and improved Dice and IOU scores. Overall, the model showed robust performance and was effective for wound image segmentation.



Classification Report

Accuracy	0.97
Test Loss	0.36
Dice Coefficient	0.76
IOU Score	0.60
Precision	0.86
Recall	0.67
F1 Score	0.75



Conclusion & Future Work

This research successfully designed, implemented, and evaluated two robust computational pipelines for medical image analysis. The Random Forest classifier demonstrated high efficacy in detecting pituitary tumours and healthy brain scans, while identifying crucial boundary ambiguities within glioma cases. Concurrently, the deep learning U-Net and ResNet50 U-Net architectures successfully captured intricate wound boundaries, showing steady loss reduction and strong spatial correlation through Dice and IOU evaluations. Together, these systems establish a reliable foundation for computer-aided clinical diagnostic tools.

Future improvements and expansion of this research will focus on:

1. **Advanced Architectural Enhancements:** Integrating transformer-based blocks into both the classification and segmentation networks to capture finer global and local contextual features.
2. **Dataset Expansion and Augmentation:** Incorporating larger, multi-centric clinical datasets with diverse noise profiles to enhance model generalizability and prevent overfitting.
3. **Explainable AI and Clinical Deployment:** Implementing visual interpretation tools like Grad-CAM for the tumour classifier and deploying the wound segmentation model into an interactive, real-time clinical web application.

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